

Project 3

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May 2026

1 Introduction

In this project, I will go over the AI Modeling method of standardization/scaling. We will discuss its strengths and weaknesses by discussing its different types and using it on different types of dataset. The main dataset used in this project is "Estimation of Obesity Levels Based On Eating Habits and Physical Condition" which is used to show the upsides and downsides of standardization. This is in addition to a synthetic dataset to illustrate the same claims.

2 Explanation of Standardization/Scaling

The method of standardization puts all features on the same measurement scale to prevent some from overdominating others. This is especially useful in datasets where the values for certain variables have a much greater range than values for other variables. For example, in this dataset about obesity, weight ranges from 39-173 kilos while number of main meals varies from 1 to 4. Without standardization, weight would dominate as it is magnitudes larger than the other variables. However, the other variables are still important which is why we need to standardize the features.

Standardization and Scaling is especially useful for classifiers that rely on gradient descent or distance measurements. Using these classifiers we don't want features of large magnitudes to disproportionately affect the model. Specifically, standardization centers the data around 0 while making the standard deviation 1 and removing unit differences in order to prevent dominance.

2.1 Types of Standardization

Additionally, there are different types of standardization and scaling methods with some more effective than others in certain use cases. The most common is typical standardization, but there is also min-max scaling and robust scaling.

Typical standardization, also known as z-score scaling, is best used for distance-based models. For example, K-nearest-neighbors on the obesity dataset. This is because it makes the mean 0 and standard deviation 1 so that the values

are in the same range as each other, making distance calculations not disproportionately affected by certain numbers.

Min-max scaling rescales everything between 0-1 and is best used when the data has known bounds. For example, the pixel values have the bounds from 0-255 which makes it hypothetically a good use case. Min-max is good for bounded range and for preserving distributions but is very sensitive to outliers.

Robust scaling uses median and IQR which centers around the median causing stable scaling while ignoring outliers. Robust scaling is best used when outliers are present or the distribution is skewed. This is good with a lot of real-world data such as income and medical among many other types.

In short, use typical standardization if you don't know which one to use, use min-max for bounded data/neural nets, and use robust scaling when outliers are present in the dataset.

2.2 Mathematical Formulation

Each standardization type has a different mathematical formulation despite the similar results.

Typical standardization or z-score scaling is done by finding the z-score of the data points. Z score is found by finding the distance between a point and the mean and then dividing it by the standard deviation. Doing this centers the data around 0 and rescales the data so that each standard deviation is equal to 1. The effect of z-score scaling is measuring how many standard deviations a point is away from the mean effectively scaling all features to the same scale.

Min-Max scaling uses normalization in its formula for standardization. The equation for it is the normalized point is shown in the figure below where $x_{\min}$ is the smallest value in the feature and $x_{\max}$ is the largest.

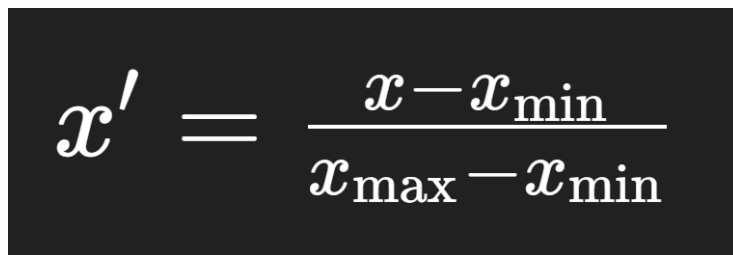
The equation for Min-Max scaling is displayed in white text on a black rectangular background. The equation is:
$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

Figure 1: Min-Max scaling mathematical formulation

This equation rescales the data to between 0 and 1. Additionally, it preserves relative distances while compressing everything to the fixed range.

Robust scaling uses median instead of mean. The equation is the point minus the median of the feature divided by the IQR. IQR being the interquartile range between Q3 and Q1. Using median in the equation is the reason why robust scaling is resistant to outliers unlike the other standardization strategies.

Standardization transforms features to have zero mean and unit variance, ensuring that all variables contribute equally to distance-based models. In contrast, min-max scaling rescales features to a fixed range, while robust scaling uses median and interquartile range to reduce sensitivity to outliers.

3 Synthetic Dataset

As said earlier, standardization is best used when the important features of a dataset have different ranges. In this synthetic dataset, I modeled uniformly random points where one feature was between 0-10000 and the other feature was 0-1. The figure below shows what the data point look like before and after standardization.

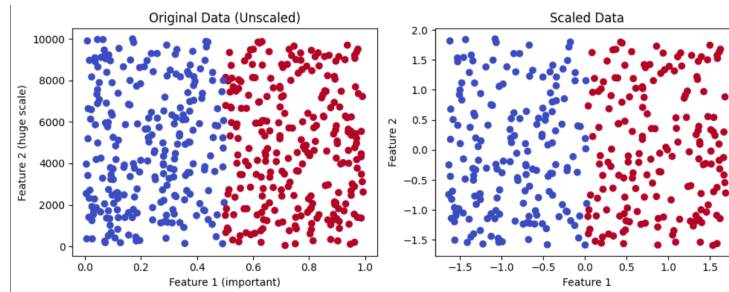


Figure 2: Visualization of data points in synthetic dataset

As shown with the graphic, the points are made to similar scales so that they are more easily comparable and one feature does not over-dominate the other.

Using typical standardization, we then looked at the accuracy of the dataset before and after. With the use of the method, the accuracy of predicting went from 0.62 to 0.97 which is very significant. Before standardization, the feature of the very large scale dominated the distance calculations. Once standardized, using KNN, the model was able to more accurately predict the desired outcome as the distance calculations were not as skewed when the ranges were similar.

This synthetic dataset showed the positives of standardization, however it is not always helpful as shown with this next synthetic dataset. This next one I made the data point closer at the start with scales of 0-2 and 0-1 respectively. Doing so made it where there was no change in the accuracy before and after standardization.

4 Real-World Dataset

We now are able to apply the method of standardization to a real-world dataset after testing it on the synthetic dataset. The dataset "Estimation of Obesity

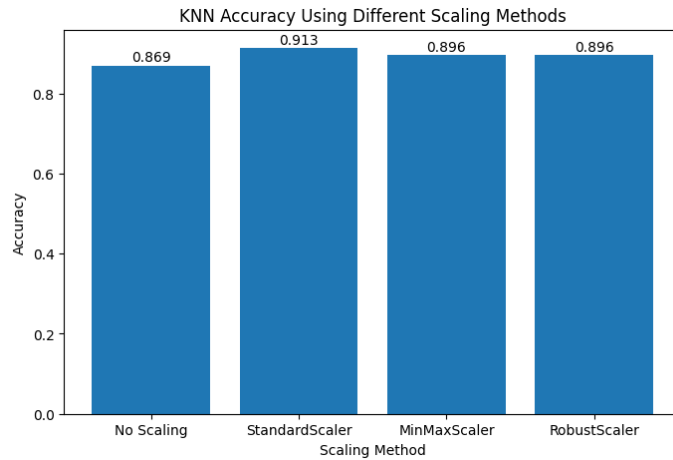


Figure 3: KNN Accuracy Using Different Scaling Methods

Levels Based On Eating Habits and Physical Condition” has 16 different variables each with different ranges. The goal of the model was to use KNN in order to predict a binary outcome on obesity based on the features of the dataset. In this section, I will show the use of standardization and the differences of the three types.

Before any method was applied to the dataset, KNN was able to accurately predict obesity based on the features with weight removed with an accuracy of 0.8691.

Figure 4 below shows what standardization does to data visually. The shape of the data distribution is maintained while the scales for the x and y axes are made to similar scales. As shown, the data is centered around 0 on both axes. This helps with KNN as the distance measurements are not overly reliant on larger features such as age.

Figure 5 shows how each of the three standardization techniques changes the data. By keeping the scale on the axes the same as before standardization, we can see how the points move. As shown, the shape of the distribution is maintained in all three techniques.

Typical standardization has the data centered around 0. This means that values above the mean remain positive while values below the mean become negative as they are standardized based on the standard deviation of a point.

Min-max scaling compressed all of the values to the 0-1 range. Since some outliers were present in age, the points were compressed more tightly together in order to maintain the distances.

Robust scaling has the data centered around the median and remains more spread out as it doesn’t rely on mean and is resistant to outliers.

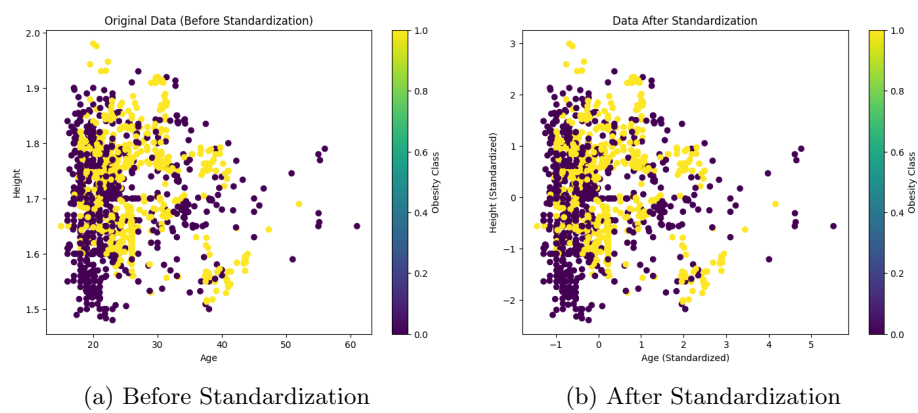


Figure 4: Comparison of data before and after standardization

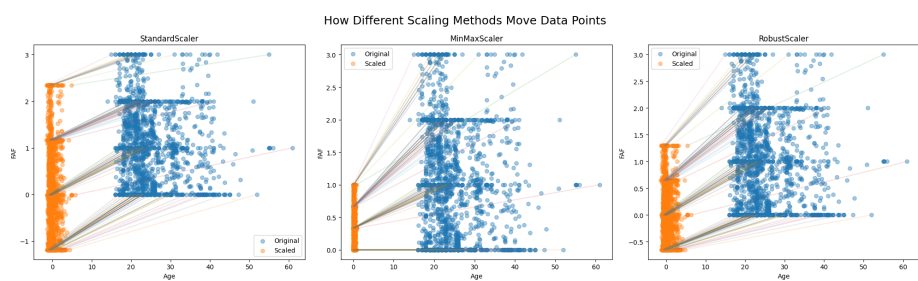


Figure 5: Visualization of Standardization Techniques

4.1 Typical Standardization

Typical Standardization improved the accuracy of the dataset giving an accuracy of 0.913. This shows how typical standardization is best used for distance-based models such as this example with KNN. It also is effective because the different features in the obesity dataset have many different ranges. For example, age ranges from around 15-60 while physical activity is only from 0-3. Standardization evens out the influence of all the features allowing for the distance metric to consider them all without the larger ones overdominating. This is why the accuracy went up significantly with typical standardization.

4.2 Min-Max Scaling

Min-max scaling although it improved the accuracy to 0.896, it was not as effective as typical standardization. Min-max scaling improved the accuracy since it rescaled the data points to be between 0-1 which made the distance calculations better. However, it was not as effective due to its sensitivity to outliers and uneven distributions which was present in some features. Outliers are bad because it means that the rest of the data is condensed to the extremes which reduces the separation between operations and thus making KNN less effective.

4.3 Robust Scaling

Similar to min-max scaling, robust scaling improved the accuracy of the model to 0.896 but was not as effective as typical standardization. Robust scaling is resistant to outliers because of its use of median instead of mean. However, there were not many extreme outliers present so its advantages were limited which is why it underperformed typical standardization.

5 Conclusion

As shown, standardization is a useful method for data analysis in order to transform observations into something readable. It is especially useful for distance-based calculations and on datasets where the features have different ranges.

We all saw the three types of standardization/scaling and their use cases—typical standardization, min-max scaling, and robust scaling. When making data useable, it is important to know if standardization is effective and specifically which type should be used.

6 Works Cited

Estimation of Obesity Levels Based on Eating Habits and Physical Condition. UCI Machine Learning Repository, University of California, Irvine, School of Information and Computer Sciences, 2019, <https://archive.ics.uci.edu/dataset/544>

/estimation+of+obesity+levels+based+on+eating+habits+and+physical+condition
. Accessed 7 Apr. 2026.

7 Acknowledgment of Assistance

ChatGPT. Assistance given to author. I prompted ChatGPT throughout the assignment for the coding and explanations but I had the ideas of what to do and add. <https://chatgpt.com/share/6a01cee7-0b94-83ea-8906-e3bb75bb28>. West Point, NY. 11MAY2026.